Nawazish Hussain

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RESEARCH PROFILE

AI researcher focused on explainable multimodal learning and scientific information extraction. My M.S. research developed an automated pipeline for extracting and tracing scientific claims across brain tumor AI literature, which led to three orally presented conference papers (AIST 2026, MNSK 2026). I also released a public dataset of 17,304 annotated sentences for claim detection. I currently develop interpretable image–text (CLIP) models for emotion recognition in visual art at National Taichung University of Education. I am seeking a PhD position in Computer Science and related fields.

Research interests: Multimodal Learning, Explainable and Trustworthy AI, Computer Vision, Information Extraction from Scientific Text, Biomedical NLP.

EDUCATION

M.S. in Artificial Intelligence and Big Data Analytics Sep 2024 – Jul 2026
Novosibirsk State University, Novosibirsk, Russia

- CGPA: **4.5/5.0**
- Thesis: *Tracing the Evolution of Scientific Claims in Brain Tumor AI: Automated Claim Extraction from Literature*
- Advisor: Dr. Evgeniy Pavlovskiy, Associate Professor

B.S. in Computer Science Oct 2018 – Aug 2022
PMAS Arid Agriculture University, Rawalpindi, Pakistan

- CGPA: **3.12/4.0**

AWARDS AND SCHOLARSHIPS

- **TEEP Research Internship**, National Taichung University of Education, Taiwan 2026
- **Open Doors Olympiad** – Winner (Computer and Data Science); fully funded M.S. at NSU 2024

PUBLICATIONS AND RESEARCH OUTPUTS

All papers accepted and presented orally; DOIs pending.

1. **Hussain, N.**, Haider, T., Pavlovskiy, E. N. *ClaimGraph-BT: Automated Claim Extraction and Cross-Paper Comparison for Brain Tumor AI Literature*. International Conference on Analysis of Images, Social Networks and Texts (**AIST 2026**), Oct. 2026. Oral presentation.
2. Haider, T., **Hussain, N.**, Zaidi, S. M. I., Pavlovskiy, E. N. *MedSETNER: A Benchmark Corpus for Extracting Dataset Names from Medical Scientific Literature*. International Conference on Analysis of Images, Social Networks and Texts (**AIST 2026**), Oct. 2026. Oral presentation.
3. **Hussain, N.** *Automatic Identification of Main Scientific Claims in Brain Tumor Research*. International Scientific Student Conference (**MNSK 2026**), Novosibirsk, Russia, Apr. 2026. Oral presentation.

Research Dataset
Hussain, N., Haider, T. *BrainTumorClaims: A Dataset for Scientific Claim Detection and Structured Information Extraction in Brain Tumor Literature*. Zenodo, 2026. 17,304 annotated sentences from 1,585 papers. DOI: `10.5281/zenodo.20364566`.

RESEARCH EXPERIENCE

Research Intern, Multimodal AI & Explainable Visual Art Analysis Sep 2026 – Present
National Taichung University of Education, Taiwan

- Fine-tune ResNet50 and CLIP models on ArtEmis for artwork emotion classification.

- Develop methods to explain model predictions and relate them to aesthetic features of artworks.

Researcher, Stream Data Analytics and Machine Learning Lab Nov 2025 – Sep 2026
Novosibirsk State University, Russia

- Built **ClaimGraph-BT**, a pipeline that detects scientific claims in 1,585 brain tumor AI papers (17,304 sentences) using fine-tuned PubMedBERT and BioBERT, extracts structured fields with Flan-T5, and links claims across papers with SPECTER2 clustering and graph layers; the best model (PubMedBERT) reached 0.84 accuracy and 0.93 AUC. Presented at AIST 2026.
- Co-authored **MedSETNER**, a benchmark corpus of 691 medical articles with 9,376 annotated dataset-name mentions (Fleiss' $\kappa = 0.85$); contributed to dataset annotation. Presented at AIST 2026.
- Coordinated lab projects in machine learning and data analytics.

SELECTED PROJECTS

- **Diabetes Prediction** – Trained and compared machine learning models on clinical data (glucose, BMI, insulin, blood pressure, age and 3 other features) to predict diabetes; tracked experiments with MLflow, versioned data with DVC, and deployed the model as a Dockerised app.
- **Age and Gender Prediction** – Python application for facial age and gender estimation using DeepFace.
- **Mobile and Web Applications** – Built three Flutter apps (Smart Agriculture Chatbot, Quran App, Hajj & Umrah Guide) with Firebase, REST APIs, SQLite and Google Maps, and a PHP/MySQL e-commerce site (Gems Bazar, B.S. final-year project).

TEACHING AND INDUSTRY EXPERIENCE

IT Trainer, Government of Gilgit-Baltistan, Pakistan Apr 2024 – Sep 2024
- Taught Python through project-based exercises; mentored learners in problem solving and algorithmic thinking.

Software Developer, Softify Technologies Jun 2022 – May 2024
- Designed, tested and deployed Android applications across multiple sectors with API integration and AI-enabled features.

TECHNICAL SKILLS

Machine Learning: PyTorch / TensorFlow, Hugging Face Transformers, scikit-learn, ResNet, CLIP
NLP: PubMedBERT, BioBERT, BiomedBERT, Flan-T5, SPECTER2, Named Entity Recognition, Information Extraction, Text Classification
MLOps: Docker, MLflow, DVC, DagsHub, Git, Linux
Programming: Python, C++, JavaScript, PHP **App Development:** Flutter, Android

CERTIFICATIONS AND LANGUAGES

- Ethics of Artificial Intelligence – Coursera (2023)
- Intro to Analytic Thinking, Data Science, and Data Mining – Coursera (2023)
- **Languages:** English (C1), Urdu, Russian (A1)

REFERENCES

Dr. Evgeniy Pavlovskiy
Associate Professor (M.S. Thesis Advisor)
Faculty of Mechanics and Mathematics
Novosibirsk State University, Russia
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